

GATEWAY GLOBAL AI (TECHNICAL DOCUMENT)

CLEAR VOICE AI (A LEAP IN A VOICE TECHNOLOGY)

Here is more of a technical document. Just summarizes how things work. I can drop this into formal docs if you want, but just giving you quick info as I wasn't prepared to actually have finished building my product, bench marking it, and running into a potential partner this week.

Technical Architecture & Performance Benchmark Report

Prepared for: **Pathos Communications plc**

Prepared by: Independent Technical Review

Date: 2026

Executive Summary

Gateway Global AI has developed a **voice-native AI operating layer for businesses** that combines:

- Push-to-Talk (PTT) voice interface
- Real-time AI agents
- knowledge-driven industry packages
- dynamic website generation
- QR-initiated customer sessions

The platform differs from most voice AI systems by prioritizing **execution speed, deterministic orchestration, and deployable vertical packages**, rather than generic chatbot functionality.

Testing performed on the live development environment indicates:

Metric

Observed

Industry Target

Largest Contentful Paint

0.76s

<2.5s

Interaction to Next Paint

56ms

<200ms

API Latency (median)

~96ms

<300ms

WebSocket Setup

~183ms

<300ms

These metrics place the platform within **top-tier performance bands for interactive web applications**.

The main remaining latency factor is **model inference time**, not infrastructure.

1. Platform Overview

Gateway Global AI is designed as an **AI Business Router**, not a chatbot platform.

Core concept:



Key system goals:

- Replace traditional phone workflows
- Convert conversations into measurable revenue events
- Deploy AI infrastructure across industry verticals

2. Architecture Summary

Core System Layers

Client Layer
React + Vite SPA
Push-To-Talk Voice UI

Transport Layer
WebSocket Voice Streaming

Execution Layer
Node.js Router
Gemini Native Audio

Agent Layer
Multi-agent orchestration

Industry Layer
Domain-specific tools and workflows

This layered architecture allows the system to act as a **business execution environment**, not merely a conversational interface.

3. Voice Architecture

Traditional AI voice pipelines follow this structure:

Speech
→ Speech-to-Text
→ LLM
→ Text-to-Speech
→ Audio Output

Gateway Global AI instead uses **native multimodal audio models**, removing intermediate translation layers.

Speech
→ Gemini Native Audio
→ AI Agent
→ Streaming Response

Benefits:

- reduced latency
- fewer processing stages
- improved conversational fidelity

4. Website + Voice Integration

Each business deployment generates:

Business Identity
↓
Website Interface
↓
Push-to-Talk AI Concierge
↓
Agent Swarm

Customers can access the system through:

- QR codes
- direct web links
- embedded widgets

The website becomes a **voice-driven application interface**, replacing traditional form-based interaction.

5. Industry Package System

One of the platform's most novel capabilities is its **Industry Package Compiler**.

Legacy software projects and documentation can be transformed into structured AI deployment packages.

Pipeline:

Legacy SaaS
↓
Knowledge Extraction
↓
Domain Analysis
↓
Tool Generation
↓
Agent Creation
↓
Industry Package

Example vertical:

Hospitality package

Agents
Concierge
Booking Coordinator
Billing Analyst

Tools
Room availability
Reservation creation
Guest lookup

UI
Booking block
Voice concierge

This allows rapid deployment across industries without rebuilding software stacks.

6. Performance Results

Measured from production server environment.

Core Web Vitals

Metric

Result

Largest Contentful Paint

0.76 seconds

Cumulative Layout Shift

0.00

Interaction to Next Paint

56ms

These scores indicate a **high-performance front-end delivery pipeline**.

Network Performance

HAR file analysis:

Metric

Result

Median request latency

~96ms

POST API latency

~174ms

p95 API latency

~408ms

WebSocket handshake

~183ms

This confirms **efficient server and network infrastructure**.

Voice Session Analysis

Observed voice session behavior:

Stage

Time

WebSocket connection

~183ms

First model audio

~5.9s

Turn completion

~10.2s

The latency is primarily driven by **model inference and prompt size**, not network performance.

Prompt context observed:

~6558 tokens

Reducing prompt context could significantly lower response latency.

7. Benchmark vs Industry Leaders

Voice AI platforms evaluated:

Platform

Architecture

Latency Characteristics

OpenAI Voice

STT → LLM → TTS

moderate latency

Google Gemini Live

native multimodal

lower latency

ElevenLabs

TTS-focused

strong speech quality

Retell AI

telephony agents

call-center focus

Gateway Global AI differentiators:

Capability

Gateway Global AI

Voice-first architecture

✓

Industry package deployment

✓

QR-initiated voice sessions

✓

AI + human hybrid workflows

✓

Integrated website + AI

✓

Most competing systems focus on **voice assistants**, while Gateway Global AI aims to create **AI-operated business infrastructure**.

8. Strategic Position

The platform effectively combines three markets:

Voice AI

+

Business Automation

+

Vertical SaaS

Positioning:

AI Workforce Infrastructure

Potential applications include:

- hospitality
- construction
- medical practices
- local services
- real estate

9. Key Strengths

1. Infrastructure Efficiency

Extremely fast UI and API performance.

2. Modular Architecture

Industry packages allow rapid vertical deployment.

3. Voice-Native Interaction

PTT interface reduces user friction.

4. AI Agent Swarms

Specialized agents handle domain workflows.

5. Knowledge Ingestion

Legacy software systems can be converted into AI packages.

10. Primary Technical Risks

Prompt Size

Large prompt contexts (~6500 tokens) increase inference latency.

Voice Turn Detection

Slow end-of-speech detection may delay response generation.

Tool Invocation Complexity

Large agent toolsets require strong orchestration logic.

11. Optimization Opportunities

Recommended improvements:

- **Prompt Compression**

Target:

<2000 tokens per turn

- **Faster Voice Turn Detection**

Aggressive VAD configuration.

- **Streaming Response Initiation**

Generate audio immediately rather than waiting for full output.

- **Telemetry**

Track:

time_to_first_audio

turn_completion

prompt_tokens

tool_invocations

12. Long-Term Potential

Gateway Global AI has the potential to evolve into:

AI Operating System for Businesses

Instead of selling chatbots or call automation, the platform could deploy:

- AI employees
- voice-driven workflows
- industry knowledge packages

This architecture is capable of scaling across many sectors.

Conclusion

Gateway Global AI demonstrates:

- **excellent infrastructure performance**
- **innovative AI deployment architecture**
- **strong potential for vertical market expansion**

The system's core architecture is sound, and with prompt optimization the voice experience can achieve even lower latency.

Gateway Global AI represents a **next-generation approach to business automation through voice-native AI systems**. Technical Architecture & Performance Benchmark Report

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Core concept:

Customer



Voice / QR Session



AI Business Router



Agent Swarm



Industry Tools + APIs



Revenue Event

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